

COMPLEX TIME, LAPSE, AND STOKES SEMIGROUP FACTORS FROM THE HQIV NULL LATTICE: A WICK BRIDGE TO NIELSEN’S COMPLEX-TIME NAVIER–STOKES PROGRAM

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ABSTRACT. Position. There is no prior HQIV publication that derives complexified temporal structure from the null lattice and compares it to Jenny Lorraine Nielsen’s complex-time Navier–Stokes program [8, 7]. The thermodynamics companion [3] records discrete dissipation signs on a C_3 heat graph but explicitly disclaims 3D Navier–Stokes. The TUFT bridge [5] imports Hopf/Beltrami spectral language for mass generation, not holomorphic Leray solutions. This note closes the gap at the *algebraic* level: it derives HQIV’s time-angle / lapse / rapidity-phase stack from lattice data already present in Tier 1, proves a single-mode Wick identification with a Stokes semigroup factor, and states precisely what would be required for full equivalence with Nielsen–Semita’s solution [8].

Result (Tier I, machine-checked). On shell index m , auxiliary field ϕ , and coordinate time t ,
 (i) the HQVM lapse $N = 1 + \Phi + \phi t$ is the real horizon term plus Newtonian potential;
 (ii) complex time $t_C = t + i \phi t$ splits real clock from time angle;
 (iii) lattice phase $\exp(i \theta(m))$ with $\theta = \phi t \delta \theta'(m)$ is the *star/conjugate* partner of the Stokes factor $\exp(-\nu k(m)^2 t_C)$ on the imaginary-time ray $t_C = i \theta / (\nu k(m)^2)$;
 (iv) for real $t > 0$, that Stokes factor has norm strictly below unity (semigroup damping).

These items are packaged in [Hqiv/Physics/ComplexTimeStokesWickBridge.lean](#).

Honest scope. This paper does **not** claim global smoothness, Millennium Option A/C, or that HQIV “solves” classical Navier–Stokes regularity—HQIV’s ontology requires the *discrete null lattice* to arrive at numbers, and both TUFT and HQIV treat finite Hopf shells as fundamental. It **does** prove (Tier I) that the HQIV lapse-modified fluid PDE reduces to the TUFT Beltrami–NS form on chart-mapped modes, and that complex-time Stokes factors are Wick-conjugate partners of lattice phase data.

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in **HQIV_LEAN**—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics.

In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. INTRODUCTION

1.1. What existed before this note. The HQIV paper series already contains the ingredients of a complex-time story, scattered across several companions:

- **Null lattice and $\phi(m)$:** the growth and action papers [4, 2] fix shell scalars and variational closure;
- **Time angle in the lapse:** `Hqiv/Geometry/HQVMetric.lean` defines $\delta\theta'_{\text{time}} = \phi t$ as the horizon contribution to the ADM lapse;
- **Rapidity phase in the zeta layer:** `Hqiv/Physics/OctonionicZeta.lean` and `Hqiv/Physics/RapidityZetaPhaseBridge.lean` identify the zeta exponent with i polarAngleFromRapidity(ϕ, t, m);
- **Effective fluid closure:** `Hqiv/Physics/HQIVFluidClosureScaffold.lean` and `Hqiv/Physics/HQIVTurbulenceSimulatorScaffold.lean` package lapse-modified momentum with eddy viscosity driven by $|\dot{\delta}\theta'|$;
- **Hopf/Beltrami mining:** [5] aligns TUFT contact spectra with HQIV shell readouts [7].

None of these manuscripts, however, states the complex-time/Stokes bridge as a single derived chain, nor compares it to Nielsen’s standalone Navier–Stokes work [8, 6].

1.2. What Nielsen–Semita claim (external benchmark). Nielsen and Semita [8] develop 3D incompressible Navier–Stokes on $\mathbb{R}^3$ using the Hopf fibration $S^1 \rightarrow S^3 \rightarrow \mathbb{CP}^1$ as an analytic tool. Their program has two temporal channels:

- **Real time:** spectral irreducibility of fiber modes $\{m^2\}$ and base modes $\{k(k+1)\}$ drives a finite-time cascade (Option C in Fefferman’s formulation);
- **Complex time** with $\Re t > 0$: Stokes semigroup damping $\exp(-\nu|k|^2 t)$ suppresses the cascade and yields global holomorphic regularity (partial Option A), with real-time smoothness recovered away from an exceptional set on the real axis.

Related work by Nielsen on Beltrami Navier–Stokes in complex time [6] sits in the same research line.

1.3. What this paper adds. We derive HQIV’s temporal structure from the lattice upward, prove the Wick bridge in Lean, and show *functional PDE identity* with TUFT/Beltrami Navier–Stokes on finite Hopf modes—without claiming continuum global smoothness.

2. LATTICE DERIVATION OF TIME ANGLE AND PHASE

2.1. Shell temperature and auxiliary field. On the HQIV temperature ladder [3, 1],

$$(1) \quad T(m) = \frac{1}{m+1}, \quad \phi(m) = \frac{2}{T(m)} = 2(m+1).$$

The auxiliary scalar ϕ is not a fitted transport coefficient: it is fixed by shell combinatorics shared with O–Maxwell and HQVM sectors [2].

2.2. Time angle as lapse horizon term.

Definition 2.1 (HQVM time angle). *For shell-level $\phi > 0$ and coordinate time $t \in \mathbb{R}$,*

$$(2) \quad \delta\theta'_{\text{time}}(\phi, t) := \phi t.$$

The ADM lapse imposed by the informational-energy axiom is

$$(3) \quad N(\Phi, \phi, t) = 1 + \Phi + \delta\theta'_{\text{time}}(\phi, t),$$

with Newtonian potential Φ .

Proposition 2.2 (Lapse decomposition (proved)). *$N(\Phi, \phi, t) = 1 + \Phi + \text{timeAngle}(\phi, t)$, and $\text{timeAngle}(\phi, 0) = 0$.*

The module [Hqiv/Geometry/HQVMetric.lean](#) proves monotonicity $\partial_t \text{timeAngle} = \phi > 0$ and the first-period identity $\text{timeAngle}(\phi, 2\pi/\phi) = 2\pi$: time angle is read mod 2π as horizon phase accumulation, not as an extra fitted parameter.

2.3. Tipping angle and polar angle on shells. The Maxwell tipping channel defines, for electric-energy proxy E' ,

$$(4) \quad \delta\theta'(E') = \arctan(E') \cdot \frac{\pi}{2}.$$

On shell index m , the *polar angle* used in the zeta layer is

$$(5) \quad \theta(m) := \text{polarAngleFromRapidity}(\phi, t, m) = \phi t \delta\theta'(m).$$

The bridge [Hqiv/Physics/RapidityZetaPhaseBridge.lean](#) proves

$$(6) \quad i \phi t \delta\theta'(m) = i \theta(m)$$

as an exponent in $\mathbb{C}$, i.e. the zeta phase factor and the polar scaffold are the same object.

2.4. Complexified lattice time.

Definition 2.3 (HQIV complex time (formal split)).

$$(7) \quad t_{\mathbb{C}}(\phi, t) := t + i \text{timeAngle}(\phi, t) = t + i \phi t.$$

Proposition 2.4 (Real and imaginary parts (proved)). *$\Re t_{\mathbb{C}} = t$ and $\Im t_{\mathbb{C}} = \phi t$. Moreover $\Im t_{\mathbb{C}} + 1 + \Phi = N(\Phi, \phi, t)$.*

Thus HQIV already carries a complex-time *coordinate*: real part = coordinate clock; imaginary part = time angle entering the lapse.

2.5. Rapidity-modulated zeta phase. The shell term in the HQIV lattice zeta [2] includes

$$(8) \quad \mathcal{Z}_m(\delta, \phi, t, s) = \text{eff}(\delta, m)^{-s} \exp(i \phi t \delta\theta'(m)).$$

The phase factor is unit modulus: it is oscillatory data on the discrete shell line, not dissipative by itself.

3. EFFECTIVE FLUID CLOSURE FROM THE SAME SCALARS

The modified fluid scaffold [2] reuses ϕ , Θ_{local} , and $\dot{\delta}\theta'$:

$$(9) \quad f(a_{\text{loc}}, \phi) = \frac{a_{\text{loc}}}{a_{\text{loc}} + \phi/6},$$

$$(10) \quad \mathbf{g}_{\text{vac}} = -\frac{\gamma}{6} \nabla(\phi \dot{\delta}\theta'),$$

$$(11) \quad \nu_{\text{eddy}} = \gamma \Theta_{\text{local}} |\dot{\delta}\theta'| \ell_{\text{coh}}^2 C.$$

Lapse-modified momentum (DNS/RANS scaffold) multiplies inertia by N from (3) [2].

Remark 3.1 (Two dissipation channels). *The phase channel (8) is oscillatory ($|\exp(i\theta)| = 1$). The eddy-viscosity channel (11) is dissipative in real time through $|\dot{\delta}\theta'|$. Nielsen's complex-time regularity uses semigroup decay on $\Re t > 0$; HQIV routes analogous dissipation through (11), not through the zeta phase.*

4. SINGLE-MODE STOKES FACTOR AND THE WICK BRIDGE

4.1. Stokes semigroup factor.

Definition 4.1 (Mode Stokes factor). *For viscosity $\nu > 0$, mode eigenvalue $k^2 > 0$, and complex time $t_{\mathbb{C}} \in \mathbb{C}$,*

$$(12) \quad S_{\nu}(k^2, t_{\mathbb{C}}) := \exp(-\nu k^2 t_{\mathbb{C}}).$$

On the real axis with $t > 0$, $|S_{\nu}(k^2, t)| < 1$: strict damping.

This is the local building block of Nielsen's complex-time Stokes semigroup analysis [8].

4.2. HQIV phase factor.

Definition 4.2 (HQIV shell phase factor). *For real angle θ ,*

$$(13) \quad P(\theta) := \exp(i\theta).$$

On shell m , $P(\theta(m))$ is the zeta phase in (8).

4.3. Main Wick theorem (proved).

Theorem 4.3 (Wick bridge / conjugate identification). *Fix $\nu k^2 \neq 0$ and real θ . Define imaginary Stokes time*

$$(14) \quad t_{\mathbb{C}} = \frac{i\theta}{\nu k^2}.$$

Then

$$(15) \quad S_{\nu}(k^2, t_{\mathbb{C}}) = \overline{P(\theta)} = \star P(\theta),$$

where $\star$ is complex conjugation. Equivalently, $P(\theta) = \star S_{\nu}(k^2, t_{\mathbb{C}})$.

Proof. Machine-checked as `stokesModeFactor_imaginaryTime_eq_hqivPhaseFactor_star` and `hqivShellPhaseFactor_eq_stokes_star` in [Hqiv/Physics/ComplexTimeStokesWickBridge.lean](#). $\square$

[Shell specialization] For each shell m with $\theta(m) = \phi t \delta\theta'(m)$ and mode scale $k(m)^2 > 0$,

$$(16) \quad \exp(i\phi t \delta\theta'(m)) = \star \exp\left(-\nu k(m)^2 \cdot \frac{i\theta(m)}{\nu k(m)^2}\right).$$

Theorem 4.4 (Real-axis damping (proved)). *If $\nu, k^2, t > 0$ are real, then $\|S_{\nu}(k^2, t)\| < 1$.*

Proof. `stokesModeFactor_pos_real_lt_one` in [Hqiv/Physics/ComplexTimeStokesWickBridge.lean](#). $\square$

5. FUNCTIONAL PDE IDENTITY WITH TUFT BELTRAMI-NS

5.1. **Shared finite-mode ontology.** Both programs reject a literal continuum as fundamental:

- **TUFT** [7]: finite Hopf shells $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$, contact Beltrami spectrum, zeta-regularized sector data;
- **HQIV**: discrete null-lattice shells $m \in \mathbb{N}$, lock-in horizon reference $M = 4$, temperature ladder $T(m) = 1/(m+1)$.

Numbers (masses, viscosities, mode budgets) are *lattice/shell readouts*. Global smoothness of classical 3D NS is therefore **not** the equivalence target; **operator-form coincidence** on matched modes is.

5.2. **TUFT Beltrami-NS momentum form.** On a Beltrami field ($\nabla \times \mathbf{u} = \lambda \mathbf{u}$), Nielsen's incompressible NS reduces to the schematic pointwise balance (continuum chart, no existence claim)

$$(17) \quad \rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) = -\nabla p + \nu_{\text{tot}} \Delta \mathbf{u} + \mathbf{f},$$

with mode diffusivity tied to Hopf eigenvalues: fiber chart $\{m^2\}$, base chart $\{k(k+1)\}$, and TUFT minimal coexact ladder $\lambda_{\min}(n) = n+1$ on winding sectors $n = 1, 2, 3$ [7, 5].

5.3. **HQIV lapse-modified form and the classical limit.** HQIV's simulator axiom packages the modified balance [2]

$$(18) \quad N \rho f \left(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nu_{\text{tot}} \Delta \mathbf{u} + \mathbf{g}_{\text{vac}} + \mathbf{f},$$

with $N = 1 + \Phi + \phi t$, inertia $f(a_{\text{loc}}, \phi)$, vacuum source $\mathbf{g}_{\text{vac}} = -(\gamma/6) \nabla(\phi \dot{\delta} \theta')$, and $\nu_{\text{tot}} = \nu_{\text{mol}} + \nu_{\text{eddy}}$ as in §3.

Theorem 5.1 (Functional PDE identity at the Beltrami classical limit (proved)). *When $N = 1$, $f = 1$, $\mathbf{g}_{\text{vac}} = 0$, and ν_{tot} is matched, the HQIV residual (??) equals the TUFT Beltrami-NS residual (??) term-by-term.*

Proof. Machine-checked as `tuftBeltramiNS_residual_eq_hqiv_classical_limit` and `tuftBeltramiNS_component_iff_hqiv_classical_limit` in [Hqiv/Physics/TUFTBeltramiHQIVPDEBridge.lean](#). $\square$

Away from this limit, HQIV's lapse, inertia, and vacuum channels are *extra physics* from the null lattice; they collapse to TUFT's Beltrami-NS template when the chart is Beltrami-aligned and gradients vanish.

5.4. Mode eigenvalue chart and three-generation bundle.

Chart	TUFT / Nielsen NS	HQIV anchor (proved)
Fiber squares	m^2	<code>tuftHopfFiberModeEigenvalueSq</code>
Base ladder	$k(k+1)$	<code>tuftHopfBaseModeEigenvalueSq</code>
Minimal coexact	$\lambda_{\min}(n) = n+1$	<code>tuftMinimalBeltramiEigenvalue</code>
Base at $k = 1$	2	<code>tuftHopfBaseModeEigenvalueSq_eq_minimal_at_one</code>
Three windings	$n = 1, 2, 3$ integrable	<code>hopfGeneration_*_has_pde_equivalence</code>

Each integrable winding carries a `TUFTBeltramiHQIVPDEFunctionalEquivalence` record: matched $\lambda_{\min}$, Wick-conjugate shell phase, and real-axis Stokes damping—no smoothness.

6. ALIGNMENT WITH NIELSEN–SEMITA: OPERATORS, NOT REGULARITY

6.1. Structural alignment (provable now).

Channel	Nielsen–Semita / TUFT	HQIV (this note)
Ontology	Finite Hopf shells	Discrete null-lattice shells
Complex time	$t \in \mathbb{C}$, semigroup on $\Re t > 0$	$t_{\mathbb{C}} = t + i\phi t$ from lapse
Oscillatory phase	Holomorphic extension data	$P(\theta) = \exp(i\phi t \delta\theta')$
Semigroup damping	$\exp(-\nu k ^2 t)$ on $\Re t > 0$	Thm. 4.4 on real axis
Imaginary-time partner	Stokes semigroup	Thm. 4.3 (conjugate)
Momentum PDE	Beltrami–NS (??)	Lapse-RANS (??); Thm. ??
Dissipation slot	Viscosity suppresses cascade	$\nu_{\text{eddy}} \propto \dot{\delta}\theta' $
Hopf spectra	Fiber $\{m^2\}$, base $\{k(k+1)\}$	Beltrami/Hopf mining [5]

6.2. What we deliberately do not equate.

- **Global smoothness / Millennium:** Nielsen’s Option A/C regularity claims are external benchmark statements; HQIV does not reproduce them and does not need them for lattice readouts.
- **Literal eigenvalue equality:** $m^2 \neq n+1$ in general; chart maps are recorded in Lean, not forced (see `tuftFundamentalBeltrami_ne_eq_peterWeyl_one` in the TUFT bridge).
- **Full holomorphic Leray lift:** extending Thm. ?? from pointwise residuals to global holomorphic solutions remains Tier III.

6.3. Legacy Tier III bundle (holomorphic extension only).

Hypothesis bundle (Tier III) 6.1 (Tier III: holomorphic extension beyond lattice modes). *If one wishes to compare with Nielsen’s complex-time regularity program specifically, the remaining items are:*

- (H1) *holomorphic Leray semigroup on a stated domain;*
(H2) *Option A/C Millennium consequences on $\mathbb{R}^3$.*

*These are **not** required for HQIV–TUFT functional PDE identity on finite shells.*

6.4. Correct statement of equivalence.

Proposition 6.2 (Provable equivalence (single-mode Wick)). *For fixed (ν, k^2, θ) with $\nu k^2 \neq 0$, HQIV’s shell phase factor and Nielsen’s Stokes semigroup factor are **Wick-conjugate partners** on the imaginary-time ray (14). They are not identical as complex numbers unless $\theta \in \{0, \pi\}$.*

Proposition 6.3 (Structural equivalence (dissipation channel)). *When $\dot{\Theta}$ is identified with $\delta\theta'(E')$ in the fluid chart, HQIV eddy viscosity (11) is proportional to $|\dot{\Theta}|$, providing a real-time dissipation slot analogous to Nielsen’s $\Re t > 0$ damping, but **not** yet proved equal to a Navier–Stokes viscosity coefficient.*

7. DERIVATION DIAGRAM

Lattice spine (Tier I definitions). Shell index $m \rightarrow T(m)$, $\phi(m) = 2(m+1) \rightarrow$ time angle $\phi t \rightarrow$ lapse $N = 1 + \Phi + \phi t \rightarrow$ polar angle $\theta(m) = \phi t \delta\theta'(m) \rightarrow$ zeta phase $\exp(i\theta(m))$.

Wick bridge (Tier I theorem). $\exp(i\theta) = \star \exp(-\nu k^2 \cdot i\theta/(\nu k^2))$ on each mode.

Nielsen benchmark (external). Hopf decomposition $\rightarrow$ incommensurate ladders $\rightarrow$ complex-time semigroup on $\Re t > 0$.

Open (Tier III). Identify $k(m)^2$ with Hopf fiber/base spectra; lift single-mode Wick to full NS.

8. CONCLUSION

This note supplies the missing HQIV manuscript for complex time: a lattice-first derivation of time angle, lapse, and rapidity phase; machine-checked Wick and PDE-functional bridges to TUFT Beltrami–NS; and an explicit refusal to conflate lattice physics with Millennium smoothness.

Proved: conjugate identification (15); real-axis damping; lapse/time-angle split (7); TUFT Beltrami–NS residual = HQIV RANS residual at the classical limit (Thm. ??); three-generation TUFTBeltramiHQIVPDEFunctionalEquivalence bundles.

Not proved / not claimed: global smoothness; identity of full holomorphic Leray solutions; Millennium Option A/C as stated in [8].

Recommended next Lean milestones:

- (1) extend Thm. ?? with nonzero lapse as a *chart correction*, not a separate PDE class;
- (2) connect HQIVFirstPrinciplesNSBridge to TUFTBeltramiHQIVPDEFunctionalEquivalence;
- (3) Peter–Weyl chart beltramiPeterWeylEigenvalueS3 for base-mode $k(k+1)$ ladder.

LEAN MODULE MAP

Paper object	Lean anchor
$\text{timeAngle}(\phi, t) = \phi t$	HQVMetric.timeAngle
$N = 1 + \Phi + \phi t$	HQVM_lapse
$\theta(m) = \phi t \delta\theta'(m)$	polarAngleFromRapidity
Zeta phase = $\exp(i\theta)$	zetaHQIVTerm_phase_arg_eq_polarAngleFromRapidity
$t_{\mathbb{C}} = t + i\phi t$	hqivComplexTime
Stokes factor	stokesModeFactor
Wick bridge	stokesModeFactor_imaginaryTime_eq_hqivPhaseFactor_star
Real damping	stokesModeFactor_pos_real_lt_one
Eddy viscosity slot	hqivEddyViscosity_HQIV_of_delta_theta_prime
PDE functional identity	tuftBeltramiNS_residual_eq_hqiv_classical_limit
Three-generation bundle	TUFTBeltramiHQIVPDEFunctionalEquivalence
Tier III holomorphic only	ComplexTimeStokesHQIVCoincidence

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